

## LITANG, China

WMO: 562570

Lat: 29.9943N

Lon: 100.2708E

Elev: 12959

StdP: 9.00

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 5.5        | 8.8 | -17.9                         | 3.4 | 26.2 | -14.7 | 4.0 | 27.8 | 21.0                  | 37.2 | 18.1 | 40.2 | 0.6                   | 90   | 0.652 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 15.2                   | 68.1            | 48.9 | 65.8 | 48.5 | 63.8 | 47.9 | 52.2                | 62.3 | 51.3 | 61.0 | 50.4 | 59.7 | 5.9                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 48.4                            | 82.9 | 56.3 | 47.6 | 80.1 | 55.4 | 46.8 | 77.7 | 54.6 | 27.3          | 62.7 | 26.6 | 61.2 | 25.9 | 60.0 | 62.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |      |     |
| 16.0              | 13.6 | 11.7 | DB                         | -1.1 | 72.5               | 3.5 | 2.3                                                | -3.7 | 74.1       | -5.7 | 75.4       | -7.7 | 76.7       | -10.3 | 78.4 | (4) |
|                   |      |      | WB                         | -3.1 | 54.4               | 3.0 | 1.8                                                | -5.3 | 55.7       | -7.1 | 56.7       | -8.8 | 57.7       | -11.0 | 59.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 41.1   | 26.1  | 29.1  | 33.9  | 40.6  | 48.1  | 53.0  | 53.7  | 53.3  | 50.6  | 43.2  | 33.5  | 27.3 |
|                                                                                  | DBStd       | 11.37  | 5.83  | 5.41  | 4.86  | 4.79  | 5.38  | 4.51  | 3.71  | 3.63  | 4.29  | 5.50  | 4.94  | 5.58 |
|                                                                                  | HDD50       | 3713   | 740   | 586   | 500   | 283   | 100   | 22    | 11    | 13    | 43    | 217   | 495   | 703  |
|                                                                                  | HDD65       | 8724   | 1205  | 1006  | 965   | 732   | 523   | 361   | 350   | 363   | 432   | 674   | 945   | 1168 |
|                                                                                  | CDD50       | 464    | 0     | 0     | 0     | 2     | 42    | 111   | 126   | 115   | 61    | 7     | 0     | 0    |
|                                                                                  | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 4.1    | 4.7   | 5.0   | 5.2   | 4.8   | 4.7   | 4.0   | 3.4   | 3.4   | 3.5   | 3.9   | 3.6   | 3.7  |
| Precipitation                                                                    | PrecAvg     | 28.0   | 0.0   | 0.2   | 0.4   | 0.8   | 2.0   | 5.6   | 7.1   | 6.0   | 4.5   | 1.3   | 0.2   | 0.1  |
|                                                                                  | PrecMax     | 43.1   | 0.2   | 0.7   | 0.9   | 2.1   | 5.0   | 9.8   | 11.6  | 13.0  | 8.0   | 4.8   | 0.6   | 0.6  |
|                                                                                  | PrecMin     | 20.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.4   | 1.6   | 3.8   | 2.4   | 1.4   | 0.0   | 0.0   | 0.0  |
|                                                                                  | PrecStd     | 5.4    | 0.0   | 0.2   | 0.3   | 0.5   | 1.3   | 2.2   | 2.1   | 2.3   | 1.7   | 1.0   | 0.2   | 0.1  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 53.5  | 53.2  | 56.2  | 62.8  | 69.3  | 70.9  | 69.3  | 69.5  | 67.1  | 62.7  | 56.0  | 52.5 |
|                                                                                  |             | MCWB   | 31.8  | 32.0  | 35.0  | 39.3  | 45.8  | 48.7  | 51.3  | 49.8  | 48.8  | 44.0  | 34.7  | 31.9 |
|                                                                                  | 2%          | DB     | 47.5  | 48.7  | 51.7  | 58.2  | 65.5  | 68.0  | 66.5  | 66.5  | 64.0  | 59.6  | 52.1  | 47.8 |
|                                                                                  |             | MCWB   | 28.9  | 30.1  | 32.8  | 37.5  | 44.1  | 48.6  | 50.7  | 49.7  | 48.0  | 42.3  | 32.8  | 29.5 |
|                                                                                  | 5%          | DB     | 43.1  | 45.4  | 48.5  | 54.8  | 62.6  | 65.0  | 64.1  | 63.9  | 61.6  | 57.0  | 49.2  | 44.1 |
|                                                                                  |             | MCWB   | 26.8  | 28.6  | 31.2  | 36.1  | 42.9  | 48.1  | 50.3  | 49.4  | 47.3  | 41.1  | 31.8  | 27.8 |
|                                                                                  | 10%         | DB     | 39.3  | 41.9  | 45.5  | 51.9  | 59.6  | 62.3  | 61.6  | 61.3  | 59.2  | 54.1  | 46.2  | 40.7 |
|                                                                                  |             | MCWB   | 24.9  | 27.0  | 30.3  | 35.1  | 41.8  | 47.9  | 50.0  | 49.1  | 46.7  | 39.7  | 30.8  | 26.2 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 32.6  | 33.2  | 36.7  | 41.2  | 48.3  | 52.7  | 53.9  | 53.0  | 50.8  | 46.4  | 37.5  | 32.7 |
|                                                                                  |             | MCDB   | 52.2  | 49.0  | 51.1  | 57.4  | 63.2  | 64.3  | 64.4  | 63.4  | 62.1  | 58.9  | 49.4  | 50.2 |
|                                                                                  | 2%          | WB     | 29.2  | 30.8  | 34.1  | 39.1  | 46.3  | 51.3  | 52.6  | 51.7  | 49.7  | 44.7  | 34.8  | 30.1 |
|                                                                                  |             | MCDB   | 46.1  | 46.5  | 48.2  | 53.8  | 61.4  | 61.8  | 62.6  | 61.7  | 60.6  | 56.6  | 48.0  | 46.4 |
|                                                                                  | 5%          | WB     | 27.2  | 29.1  | 32.4  | 37.6  | 44.8  | 50.2  | 51.7  | 50.7  | 48.6  | 43.2  | 33.0  | 28.3 |
|                                                                                  |             | MCDB   | 42.2  | 43.7  | 45.5  | 51.0  | 58.3  | 60.4  | 61.1  | 60.4  | 58.7  | 53.8  | 46.6  | 43.2 |
|                                                                                  | 10%         | WB     | 25.1  | 27.4  | 31.0  | 36.2  | 43.5  | 49.2  | 50.8  | 49.8  | 47.6  | 41.6  | 31.3  | 26.4 |
|                                                                                  |             | MCDB   | 38.6  | 40.8  | 43.4  | 49.0  | 56.1  | 59.1  | 59.5  | 59.2  | 56.9  | 50.9  | 44.3  | 40.0 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 26.2  | 24.7  | 22.1  | 20.4  | 19.6  | 17.0  | 15.2  | 15.7  | 16.5  | 20.0  | 24.6  | 27.0 |
|                                                                                  |             | MCDBR  | 31.4  | 29.8  | 26.7  | 25.1  | 23.5  | 21.1  | 19.2  | 19.9  | 19.5  | 22.5  | 28.5  | 31.6 |
|                                                                                  |             | MCWBR  | 17.5  | 16.0  | 12.5  | 9.7   | 7.9   | 7.0   | 6.9   | 7.0   | 6.9   | 8.5   | 13.3  | 17.1 |
|                                                                                  | 5% WB       | MCDBR  | 29.7  | 27.5  | 22.9  | 21.9  | 20.0  | 16.8  | 15.3  | 16.1  | 17.0  | 18.8  | 25.8  | 30.4 |
|                                                                                  |             | MCWBR  | 17.1  | 15.1  | 11.3  | 8.8   | 7.5   | 7.0   | 6.6   | 6.7   | 6.9   | 7.6   | 12.3  | 16.6 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.201  | 0.225 | 0.273 | 0.289 | 0.276 | 0.274 | 0.286 | 0.288 | 0.262 | 0.230 | 0.206 | 0.194 |      |
|                                                                                  | taud        | 2.667  | 2.592 | 2.438 | 2.446 | 2.570 | 2.679 | 2.639 | 2.613 | 2.701 | 2.775 | 2.750 | 2.735 |      |
|                                                                                  | Ebn at Noon | 341    | 338   | 323   | 318   | 320   | 318   | 315   | 315   | 323   | 332   | 337   | 341   |      |
|                                                                                  | Edh at Noon | 23     | 28    | 35    | 36    | 32    | 29    | 30    | 30    | 27    | 23    | 21    | 21    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1313   | 1497  | 1697  | 1829  | 1870  | 1770  | 1634  | 1600  | 1509  | 1430  | 1332  | 1250  |      |
|                                                                                  | RadStd      | 66     | 94    | 128   | 106   | 99    | 151   | 135   | 166   | 96    | 85    | 82    | 58    |      |

## Historical Trends

|                        |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only           |  | +1.21 | +1.66   | -2.05  | +1.71   | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (6 neighbors) |  | +0.81 | +0.85   | -2.03  | +1.26   | N/A   | N/A   | -137        | -292  | +211  | +13   |

Nomenclature: See separate page